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Author: Edward Wilkin Perry

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BANANAS

NATURE’S INSTITUTION FOR THE PROMOTION OF LAZINESS

BY EDWARD W. PERRY

REVISED EDITION

NOTE

The chapter given in the following pages is from a work entitled: "Tropical America: Its Planters and Plantations," now in preparation. *Sports Afield* said of the author: "Probably no American is more competent to write of the country life than is this author, who, because of his long-trained habits of observation, careful search for the bottom facts and weighing of details, of deducing therefrom the essentials and presenting them clearly and concisely, has made the best possible use of his time and experience."

Perry Press
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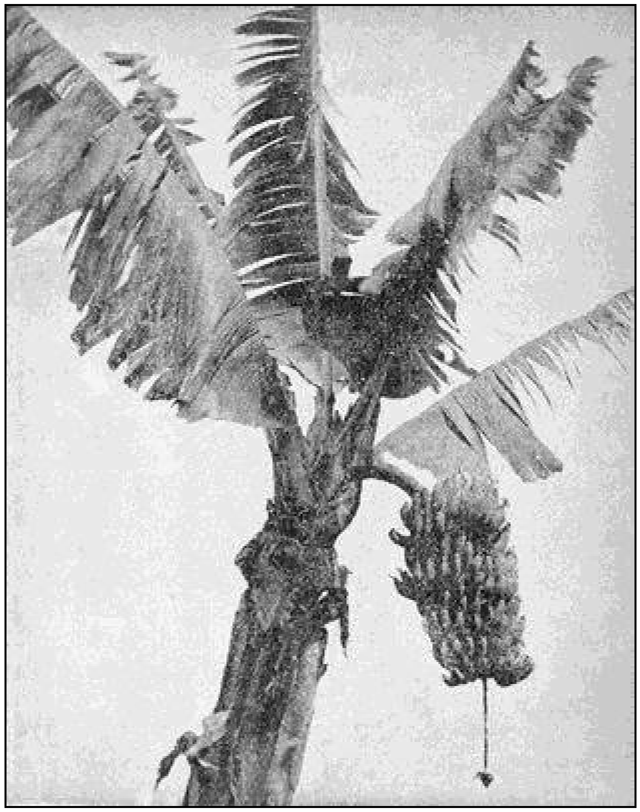
CHAPTER III.

Nature's Institution for the Promotion Of Laziness. Bananas: What they are, how they grow, what they cost, and what they give to man.

Long before the dawn of history in the Old World, mayhap long before that Old World arose from the waters, man lived on the fruit of the *Musas*. There are those who would tell you that the banana is the fruit which tempted Eve, to the downfall of Adam; and that evidence of the truth of this may be found in the fact that if one will cut across a banana, of the right kind, he may find in its heart the sign of the cross; and in the other fact that men of learning have given to a banana the name of *Musa paradisiaca*, which being interpreted means the Fruit of paradise, and to another banana they have given the name *Musa sapientum*, which the sapient know means the Fruit of knowledge. Less evidence has served well enough to burn heretics at the stake.



A BUNCH OF BANANAS

Man has carried this gigantic herb to every fertile spot in a belt that girdles the waist of the globe—a girdle that is four thousand miles and more in width. Millions uncounted have looked to it for the chief of their diet, as other millions have looked to the cereals. And to this hour puling babes and doddering ancients are fed with the fruit in all its stages and conditions, green or over-ripe, raw or roasted, baked or fried, liquid or dried. At least forty species of the *Musas* are known and described, and of these there are several sub-varieties. They have been classed by Dr. Sagot into three groups, as follows:

Giant bananas, of which *M. ensete* is the type. In this group no suckers are formed. Fruit leathery and not edible, with few seeds.

Fleshy-fruited bananas; *M. sapientum* the type. Stem produces suckers; spike long and decurved; fruit fleshy and usually eatable.

Ornamental bananas. Spike often erect, not pendant; bracts persistent, brightly colored, each with a few flowers on its axil; suckers many; fruit leathery. *M. rosacea* furnish familiar examples of this group.

When in the course of human events it becomes necessary for the single man of the tropics to take unto himself a helpmeet for him, and to provide for other events likely to come after, he selects some fertile spot, usually on the border of waters over which his canoe may easily carry the bulky harvests he will have; and there he cuts down tree and vine, bush and bamboo, and lets them lie as they fall in tangled mass. Every day the ardent sun helps the constant wind to shrivel leaf and twig until, one day, the windward edge of that snarl is touched by the torch, and in a moment a blazing hades is where a cool and shady grove will soon rustle in the breeze.

When the last flame has flickered out and coals lie dead beneath their gray shroud, women paddle to that place with canoes laden with banana sprouts. With machetes they dig little pits amid charred stumps and trunks and branches, and in each hole they set a sprout. Then they go away to wait, and rest; and the sun shines warmly down into that clearing, breezes sift a gray veil of ashes over the wilted suckers that look like black and ragged stakes; and at last come showers which wash them clean.

Those stakes are made up of sheathes of leaves tightly rolled one around another, the inner ones narrow, cream-colored and tender; those nearer the outer ones wider and yet wider, until the outer one is reached. The outer one covers nearly or quite three-fourths of the stem. When the warm rains fall, the tender leaves unroll and spread to their widest, and the sun dries and the wind whips them until soon they are split into narrow ribbons; and a few weeks after that planting a sea of giant leaves waves and whispers in the breeze—a roof of bright and tender green covering the moist, black ground.

Not before the plant has grown to a height of ten to twenty, and in some places to thirty feet, does the flower-stem begin pushing its way up from the base through the middle of the stalk. In a short time it sends out at the top one or more leaves, smaller than their older fellows, as a signal that flower and fruit will quickly follow. Soon every supporting column of those graceful arches ends in a cone of red that deepens into purple and swells until its outer petals are crowded off by the fatness of the fruit they hide, that these may have air and light. Under those petals the baby bananas are packed close, like fingers tightly gripping the parent stem. These closed ranks, each separate hand or whorl reaching half way around the stalk, grow so quickly that in six or eight weeks the bunch weighs fifty pounds or more.

To most people of northern climes bananas are merely—bananas. For such folk know as little of the many varieties of bananas as they know of the many and varied uses of that fruit. Perchance that is why they fry the common yellow guineo which comes by millions of bunches each year to the United States, and then wonder that folk who have dwelt in the tropics, and who extol fried bananas, show nevertheless that they cannot like the mushy, cloying mess set before them here. He who grows bananas, and she who cooks them for him, select for frying that thick-bodied, hard-fleshed and rather tart fruit which they call plátano, and which is by blundering English-speaking tongues misnamed plantain. And even among the plátanos there is room for choosing, for there are of them several varieties. Best of these is that little one which bears, on the Mosquito Shore whence good bananas come, the Spanish name “miel,” or honey, coupled with the Waika word “silpe,” or little. The name “maiden” plátano also is given to the “little honey,” most fittingly, for it has just enough of piquant tartness to give unfailing relish, yet is tender, plump and mighty comforting withal, upon occasion.

If he is so lucky as to live near a port where steamships stop, the planter may sell his plátanos for a cent or even two cents for each finger or fruit; and as the plants may be set only eight or ten feet apart, and each will mature a bunch of thirty to fifty fingers every nine months, it is clear that he who has an acre of plátanos may have a tidy income of food or of cash. Usually the planter prefers to eat this food, for which reason people in the North have few opportunities for learning the superior virtues of the fruit. The planter is quite right, for the plátano is the one banana fit to be cooked; and is by no means bad to eat raw.

Sometimes a planter may leave a bunch of bananas to ripen on the standing stalk, but that will rarely be, for the fruit so ripened is strong in flavor, dry and too soft to bear transportation; its skin splits, and ants, bees and other insects gather about the exposed flesh. Therefore the women lug home green bunches and hang them in the house to ripen, where everybody who has the right—and that is every visitor, every member of the family and every passing acquaintance—may pluck and eat as the fruit turns yellow and becomes tender. Meanwhile many of the fruits will have been taken from the bunch, peeled and broken into bits, to be boiled with beef or pork, or flesh of the deer, peccary or other game.

Another sub-variety of plátanos bears, in Mosquitia, the name of "butuco," perhaps from the name of the River Patuca—or maybe the river has taken its name from the banana. The butuco is perhaps rather more tart than the miel silpe, and when fried reminds one of fried greening apples, and when stewed has somewhat of the flavor of stewed peaches. In either way it is most agreeable to the taste. There are other plátanos, also, most of them giants among bananas, many being fifteen or more inches long and some two or three inches in diameter. These are firm in flesh, resist decay much longer than do the common guineos, and will, therefore, much better bear transportation. They should become known to the millions of northern lands, for they would afford a vast supply of food much more convenient and palatable than, and equal in value to, potatoes.

Prof. Wynter Blythe, of London, is an analyst who tells us that the relative values of bananas and sago, corn meal and wheat flour are as follows:

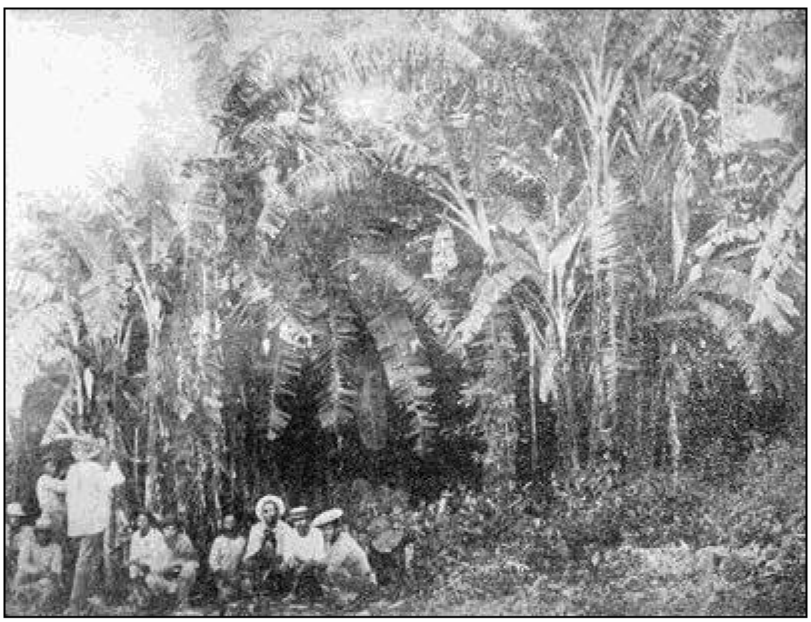
Constituents	Banana	Sago	Corn Meal	Wheat Flour
	Per Cent.	Per Cent.	Per Cent.	Per Cent.
Water	8.05	13.00	11.09	15.08
Soluble albumen dextrine	4.45			
Starch	82.57	78.06	85.30	81.60
Albumenoids	2.28	2.57	2.37	2.11
Fat	0.77			
Ash	1.88	0.53	0.43	0.35

In a report on the constituents and food values of most articles in common use on northern tables, the United States Department of Agriculture gave, in the year 1903, very valuable figures which show that nineteen vegetables and ten varieties of fruits which make up the chief of our diet, have the following parts and values:

Elements	Vegetables	Fruits	Bananas
Carbohydrates, parts	8.9	11.1	14.3
Fats	0.4	0.4	0.4

Protein	2.0	0.6	0.8
Ash	0.9	0.5	0.6
Water	73.0	64.3	48.9
Refuse	14.8	23.1	35.0
Fuel values	203.9	204.0	260.0

This shows that while of valuable nutritive elements, the nineteen fresh vegetables have 11.3 parts and the ten varieties of succulent fruits have 12.1 parts, the bananas have 15.5 parts. From this it appears, also, that if the fresh fruits and vegetables were actually worth, as food, say $1.17, bananas of like weight would be worth 38 cents more.



HARD LABOR AMONG THE BANANAS

Statements made by other analysts seem to warrant the deduction that the nutritive value of a ton of potatoes, at one cent per pound, is 19 cents more than that of a ton of bananas at the same price. There is a difference, too, in the cost of production of a ton of potatoes and the cost of raising a ton of bananas. The field for potatoes must be plowed and harrowed in the spring, the seed dropped in furrows, which are then to be covered, after which comes cultivating again and again until the time has come for digging and picking, carting, sacking and hauling, often to a distant market.

Luckily for the millions who have depended so largely on the banana for sustenance, the plant has few, if any, insect enemies and diseases, in which they differ somewhat from some fruits and tubers of the North.

Many times an assertion has been printed to the effect that Humboldt said that an acre of bananas yields forty-four times as much food as does an acre of wheat. In the year 1902 the average yield of wheat in the United States equalled 12.79 bushels, or 767.4 pounds. This had a food value equal to nearly one-third that of the average output of bananas from an acre. It is often said that one pound of bananas has as much nutrition as has a pound of beef. The truth is that one pound of beef is worth three and one-third pounds of bananas. Bananas are far enough ahead of the harvests the farmer of the North gets, without making exaggerated claims for the fruit of the tropics.

So the planter of bananas has each year four and a half times as much palatable food from an acre as the farmer gets from his potatoes: and there is the further difference that the one has bananas at no other cost than that of keeping down bush and grass and vine, that would quickly cover every spot to which the sunshine could penetrate, along the edges of the plantation. For bananas yield year after year without replanting. Each new stalk springs from the foot of its parent, grows to a height of fifteen to thirty-five feet, bears its burden of luscious fruit, and dies; but not before it has sent up from its own root new stalks to fruit and die—and so on through the centuries.

He who would grow bananas for market must plant on the border of navigable waters giving access to some harbor or anchorage where ships may safely lie while receiving the fruit. For it is easily bruised, and wetting by salt water blackens the skins, thus injuring or preventing the sale. Plantations are usually on the banks of rivers or of estuaries, but some are beside railroads, to which the fruit is carried by carts thickly carpeted with banana leaves. A cruder way is to hang a few bunches over the back of a burro or of a mule, which plods along to the shipping place.

It is evident that the entire area which can so be devoted to banana culture must be small, for most Central American and Mexican rivers are obstructed at their mouths by sandbars, over which ships cannot pass. Bluefields, Nicaragua, has been a most profitable field for banana growing, because it has a river into which sea-going ships can safely enter, and up which such ships may go fifty or sixty miles, and receive their cargoes from landings on the plantations which border the Rio Escondido. Yet millions of bunches of bananas have been shipped from the open coast of Honduras, where the one good harbor is that at Puerto Cortez.

Other millions have been shipped from Port Limón and from Bocas del Toro, in Costa Rica, whence a few hundred bunches were sent as a beginning to the United States in the year 1883. Twenty years later the port of Limón itself sent 4,174,200 bunches to the markets of the world. They brought to Costa Rica credit for producing the best bananas known.

For ages the native of banana lands was content with the fact that he got from his plantation more than enough food. Some thirty-five years ago a few bold men ventured to pay twelve or fifteen cents a bunch for a few cargoes in the Bay Islands, off the coast of Honduras, and carried them to the Gulf States. There they found they could sell the fruit, for there lived people who had traveled to the tropics, and learned to eat their foods. To-day millions of bunches are each year sold in the United States and even in Canada, and in 1902 ship-loads were sent from Costa Rica direct to Europe. That little republic alone received not less than $1,127,400 for bananas sold abroad during the year that ended with September, 1902.

The United Fruit Company, of Boston, was formed in the year 1888, and ten years later was said to have a surplus of more than $6,000,000, owned thousands of acres of bananas, and had built expressly for its fruit carrying business four superb steamers, and employed many others.

It is safe to assume that more than $6,000,000 was paid in the year 1902, in Central America alone, to planters of bananas. Nearly all of that was paid by products of American farms, factories and forests. Farmer, manufacturer and miner, lumberman, railroad man and sailor, merchant and broker of this country, are all concerned in and benefited by the work done in shady aisles beneath banana leaves on the banks of tropic rivers.

Bananas reach their best estate on the low, deep alluvium near the Caribbean coast, where the temperature never sinks below 60° and is seldom below 80° F. Such low lands serve all the better if flooded two or three times in the year, for the banana will drink much water, and such floods bring silt from the hills, and thus keep the ground fertilized without cost to the owner. In 1897 famed banana fields of the Rio Escondido were so deeply flooded that the steamship “Saga” voyaged through the main streets of Rama, fully sixty miles from the mouth of the river, to pick off from their roofs the dwellers in that town. The bananas barely showed their tops above the yellow flood. Along the coast flew reports that the plantations were ruined; subscriptions were asked to help the planters: and three months later they were harvesting better crops than in years before. Their plantations had been so enriched that they bore most bountifully.

Bananas may be grown wherever there is some moisture and no near approach to the frost line; but a touch of frost cuts down the banana as a breath from a fiery furnace would blight a tender lily. The city of Tegucigalpa is 3,600 feet above the level of the sea, yet in that town is a field some thirty feet above the current in the swift river which it borders. It is very dry during months of each year, but in that field are plátanos which reach a height of more than twenty feet and bear bunches enough comfortably to support the owner. In narrow cañon and wider valley near that place are many patches of bananas which bring to their planters a sufficient income. And at that altitude the mercury sometimes falls below 65° Fahrenheit.

In the land of bananas, cats, dogs and pigs, mules, horses and cattle, parrots, babies and all other domestic animals thrive on this perfect nature-food, when they can get it. I have seen an Indian woman pry open with her fingers the jaws of a baby peccary, and with a gruel of green bananas choke off its incessant, rasping cry of “ma, ma!” And the next instant she put that same calabash of gruel to the lips of her own babe of three or four months. I’ve seen other Indians feed infant tapir, suckling jaguar, skinny squabs of parrots and very young monkeys on such pap, which those folk call wabool. I, myself, have safely carried abandoned cardinals through from their infant days of a beggarly few pin feathers to those of full regimentals of brilliant scarlet and epaulets of jet; and they were as overflowing with joyful song and saucy happiness as they could have been had worms and bugs been the chief of their diet every day of their lives, instead of the bananas on which they had been largely fed.

Why not, indeed, when cakes and beer, brandy and sugar, pies, puddings and sauce, and many another thing good for man to take for his stomach's sake, are made from bananas. So, too, are paper and laces, brushes and cloth, and cordage enough to pull up the earth by its roots, if only we had a place to hook the tackle.

HARVESTING BANANAS

When he has set out an acre or two of bananas, the planter need have no fears for the future. He has ample insurance against such privations as come from illness, accident or old age: and they who by a little labor pay for such insurance share each day its material benefits. No need for them to die that others may enjoy the blessings of such wise provision; nor need the planter toil with hoe or spade, cultivator or plow. It may be he will slash away with machete such vine or sapling, grass or weed as happens to obstruct his path; but as a whole he interferes as little as possible with the operations of kindly Mother Nature. She is more than ready to do his work: he is willing to let her do it.

He whose acre of bananas has been well planted has on it 225 hills, or 900 stalks. Each stalk will give him a bunch which, on rich, new ground, should weigh 60 pounds, say 54,000 pounds each 12 or 14 months. That is the theory. The fact seems to be that the average yield is really 175 to 300 full bunches to the acre per annum, say a mean of 270 bunches weighing about 16,000 pounds. The average yield reported all along the Caribbean shore and from Jamaica, during a dozen years, equaled 270.95 full bunches an acre per annum.

In the year 1902 the average yield of potatoes in the United States was 80.44 bushels per acre, and the average farm value was 49 cents per bushel, or $39.45 an acre. In Costa Rica the average price of bananas on the plantation was equal to at least 27 cents a bunch. At that figure 261 bunches would bring $70.47. In August, 1903, the price was raised to 31 cents a bunch on contracts to run three to five years; which should give $84.00 per acre each year. That is a cash difference of $44.55 in favor of the man whose bananas raised themselves for him. There was another difference in his favor, for his fruit may be eaten green or ripe, raw or roasted, boiled or fried, with fish, flesh or fowl, or with none of these.

Those who dwell in the mountain regions, far from the ports whence bananas are shipped, dip in lye and dry in the sun many a plátano. It is then shriveled, moldy-looking and altogether unlovely; but if kept dry it remains sweet and wholesome many a year. It may be eaten uncooked, when it is a gummy, sugary paste; but drop it into scalding water, put it into a hot oven, or stick it up beside the fire, and it becomes mightily puffed up, tender and savory. It might be sent thus dried to feed the people of the North or of Europe, for it would be easily packed and carried.

Naturally the intelligent planter concerns himself mainly with the question: What is the cost, the yield and the profit of banana growing? There are evidences that many people in the North feel a lively curiosity about the same points.

Before one can give a trustworthy reply to such question he must study the evidence of those who have had opportunity to learn the truth, and he should be able to present the general averages of the results shown by many such witnesses. The planter of medium ability and industry may confidently expect to attain the average results; he who has less intelligence and thrift should not complain if he fails to get as good returns; he who shows more than common skill, application and energy will win greater reward than is shown by the average of the banana-growing of the many, as in other occupations great skill and industry bring the larger rewards.

Reports covering years of experience by thousands of planters in the West Indies and along the Atlantic coast of Mexico and of Central America, indicate that the cost per acre of making banana plantations and cultivating and harvesting the first crop therefrom, the yield in bunches and the income, are as shown in the following table:

Countries	Bunches	Income	Cost	Profit
Costa Rica	250.0	$ 70 67	$ 28 84	$ 41 83
Guatamala	267.5	124 36	42 80	81 56
Honduras	294.0	121 13	18 97	102 16
Jamaica	288.0	109 48	27 58	81 90
Mexico	280.0	123 61	28 12	95 49
Nicaragua	246.2	86 36	22 07	64 29
Averages	270.95	$ 105 94	$ 28 06	$ 77 87

From the foregoing it appears that the general average yield per acre during the twenty years covered by the figures given, was 270.95 bunches per acre; the average cost per acre was $28.06, which was only 10.3 cents per bunch. The profit per bunch was 28.7 cents, or 287.9 per cent.

A report dated August 1, 1903, by Las Haciendas de Santa Clara, Costa Rica, which has 550 acres of bananas in full bearing, and where wages are one colon or 47 cents per diem, gives the cost of cultivating and delivering the fruit at the railroad, as $17.69 per acre, the yield at 173 bunches and the income at $54.90 annually. That shows that the bananas cost 10.2 cents per bunch, and that the profit was 20.8 cents a bunch, or 200 per cent. But as the fruit is sold five years ahead at those figures, the small percentage of profit may be regarded as a fair return for the investment, combined as it is with an assurance of continued gain.

There are those who insist that the higher results shown in the foregoing table may easily be obtained by any one who will give as much thought and labor to growing bananas as are required for the successful raising of corn or of potatoes. It is true that the figures on which the averages shown are based were, in many cases, from the experience of native and other planters of little diligence and skill, and that they got smaller results than might easily have been obtained. It may be possible that if one will allow two or three stalks to rise from each stand of bananas, and together mature their fruit, he many get 444 to 780 bunches from an acre each of a few years, and that in such a case he might get $185 to $278 for the crop; but it will be clear to all that he who expects to make only 270 bunches per annum from an acre, and get only $78 profit therefrom, will be safer than he who invests his money with the expectation of making greater gains.

The Hand Book of Nicaragua, published by the Bureau of American Republics, which is under the direction of the U. S. Department of State, says:

> There is, perhaps, no industry in Central America that is more attractive to men of small capital than banana growing, from the fact that the clearing of the land is effected cheaply, and from the small cost of after-cultivation, which is limited only to such clearing of weeds and undergrowth as may be sufficient to allow access to the trees, and the short time necessary to produce a paying crop. When the trees and brush that have been cut in clearing the land become sufficiently dry, they are burned, and the banana suckers are then planted among the charred remains and ashes, without any further preparation of the soil. The best results are obtained by giving the trees plenty of space, say from 15 to 18 feet apart. In about ten months the first fruit can be gathered; but in the second year the trees reach maturity, and by a proper management of the fruit stalks in a fair sized plantation a constant succession in the crop may be secured, and fruit gathered every week throughout the year.
>
> The only careful work necessary on a banana plantation is in handling the heavy bunches so as to avoid bruising them, as any such injury causes a black spot to appear, beneath which decay quickly begins as the fruit ripens. The natives have learned by experience when they cut into the fruit stalk so to gauge the strength of the blow as to cut just deep enough to cause the stalk to bend slowly over until the end of the bunch reaches the ground, when another slash with the machete severs it, and it is loaded carefully into the cart.
>
> A plantation of 40 manzanas (about 69 acres) will, during and after the second year, produce about 54,000 bunches. The lowest price paid for bunches for some years past is 37½ cents per bunch, which would give an annual value of the crop of $20,250, or more than double the expenditure for purchase of land, clearing, cultivating and gathering the crop, and all expenses to the end of the second year.

As the cost of producing bananas after the first crop from a plantation is confined to cultivating and harvesting, which may be done for $10 per acre yearly, it is scarcely wonderful that Judge O’Hara, late U. S. Consul at Greytown, Nicaragua, a lawyer whose acute mind is trained to sifting evidence, reported to the Department of State at Washington regarding banana-growing on the Atlantic coast of that republic, that:

It seems reasonably certain that bananas on the Bluefields River pay better than many crops in the United States. * * * * These figures would seem to indicate that at the end of a year a planter having 36 acres of bananas under cultivation would have $3,847.32 left after paying for all necessary labor and provisions—figures apt to bring discontent to an American farmer having but 36 acres of wheat or corn; and especially so when he compares the price of his land, ranging from $15 to $80 per acre, with that of land in eastern Nicaragua, where cultivated lands may be said to have no established market value, few improved plantations having ever been sold.

BEGINNING OF A LONG JOURNEY

Such discontent might be aggravated by consideration of the differences which exist between the results obtained from the chief eight crops of the United States and those shown by the foregoing summary of banana farming. These differences are illustrated by the following figures, those for the crops of the North showing the yield and values for the year 1897. The last column shows the difference in favor of bananas per acre:

CROPS	Yield per acre	Value per acre	Difference, favor of Bananas
Barley, bushels	23.11	$12 34	$93 59
Buckwheat,	16.08	9 69	96 25
Corn,	24.62	9 51	96 43
Oats,	27.19	8 29	97 65
Potatoes,	80.44	39 45	66 49

Rye,	13.30	8 22	97 72
Wheat,	12.78	10 11	95 83
Hay, tons	1.26	10 93	95 01
Tobacco, pounds	797.30	55 81	50 13
General averages		$18 28	$87 66

From this it is evident that bananas give five and one-half times as much as the principal crops of the United States give the farmer for his toil.

Many native planters seem content with the returns their bananas give, and appear to have no thought of increasing that income.

“Why don’t you plant more bananas? See how well this little patch has paid,” I have said to many of them.

“Why should I do that? Have I not plenty to eat? I have enough money; if I plant more I shall have to do more work to get more money which I don’t need,” is the substance of their replies.

Years ago U. S. Consul Burchard complained of the banana business of the Honduras coast, that “A large proportion of the fruit-growers were formerly vacqueros in the interior, working on a salary of $30 to $40 a year. They are now owners of plantations, and have a steady income of $30 to $300 a month. The large amount of money distributed along this coast in exchange for fruit would make any civilized and temperate community prosperous and happy. There would be public and private schools, churches and banks, newspapers and libraries, parks and carriages, and handsome dwellings supplied with every comfort and luxury, surrounded by gardens of flowers, fruits and vegetables natural to this climate of perpetual seedtime and harvest.”

So it soon will be, for already Italian and German, Englishman and American have accepted the invitation of a most kindly Nature, and the sincere welcome of friendly natives, and cottages peep here and there from out the glossy greenery, hammocks swing beneath the never-ceasing rustle of the palms in the blessed trade winds, and the fruit of Paradise gives to all a most generous support.

But those who have good lands back from navigable water and remote from railroads, are not without hope of profit from bananas. For they may dry the fruit, pack it in dainty boxes with a liberal dusting of sugar to fill vacant spaces, and send it to the hungry millions of Europe. This has been successfully done by planters of Trinidad and of Jamaica, who, in at least some instances, found that they could sell the dried fruit at 16 to 20 cents a pound. Green bunches average nearly 60 pounds in weight, two-thirds of which is lost in peeling and drying, leaving about 20 pounds, which, at 15 cents, will give $3 per bunch. If the production of the green bananas and the drying should cost $2 a bunch, the income from an acre of bananas would be $288 yearly. In practice it has been found that the total cost and income of dried bananas give a net return of $2.72 per bunch, which equaled about $783 per acre.

Both plátanos and guineos, or ordinary yellow bananas, may be profitably dried or made into flour. This will utilize the surplus fruit and such bunches as are too small to sell to advantage. Frequent mention is made by Stanley, of banana flour in his "In Darkest Africa." He strongly indorses its nutritive qualities, and wonders that the natives did not appear to have discovered what invaluable nourishing and easily digested food they had in the plátano and banana. He expressed the conviction that, "If only the virtues of banana flour were publicly known, it is not to be doubted but it would be largely consumed in Europe. For infants, persons of delicate digestion, dyspeptics and those suffering from temporary derangement of the stomach, the flour properly prepared would be of universal demand. During my two attacks of gastritis a light gruel of this, mixed with milk, was the only matter that could be digested."

It is interesting to note that such a high authority as the "Dictionary of Economical Productions of India" says:

> The large crop of food produced by bananas and plantains may be preserved for an indefinite period either by drying the fruit or by preparing meal from it. When the nearly ripe fruit is cut into slices and dried in the sun, a certain part of the sugar contained in the fruit crystalizes on the surface and acts as a preservative. The slices thus prepared, if made from the finer varieties, make an excellent dessert preserve, and if from the coarser, may be used for cooking in the ordinary way. They keep well if carefully packed when dry, and ought to form a valuable antiscorbutic for long voyages. The fruit may also be similarly preserved whole by stripping off the skin and drying it in the sun. Plantain meal is prepared by stripping off the husk and reducing it to powder, and finely sifting. It is calculated that the fresh core will yield 40 per cent. of this meal, and that an acre of average quality will yield over a ton.
>
> Plantain meal is of a slightly brownish color, and has an agreeable odor, which becomes more perceptible when warm water is poured upon it, and has a considerable resemblance to that of orris root. When mixed with cold water it forms a feebly tenacious dough, more adhesive than that of

oatmeal, but much less so than that of wheaten flour. When baked on a hot plate this dough forms a cake which is agreeable to the sense of smell, and is by no means unpleasant to the taste. When boiling water is poured over the meal it is changed into a transparent jelly, having an agreeable taste and smell. Boiled with water it forms a thick gelatinous mass, very much like boiled sago in color, but possessing a peculiar pleasant odor.

In this connection it may be interesting to note that, according to an analysis published in the *American Analyst*, New York, February 15th, 1893, the chemical composition of bananas and potatoes is almost identical, as shown by the following comparison:

	Banana	Potato
Water	75.71	75.77
Albumenoids	1.71	1.79
Total carbonaceous matter (non-nitrogenous)	20.13	20.72
Woody fibre	1.74	.75
Ash	.71	.97

Nor do the food elements in bananas and plátanos vary greatly, the sum of each being about the same.

In a communication to Kew by Mr. Louis Asser, of the Hague, Holland, it was announced that a syndicate proposes to take up the manufacture of banana and plantain meal and the preparation of dried bananas on a large scale in Dutch Guiana. The communication referred to gives the following list of commercial preparations from the banana and the plátano:

1. Dried slices of the entire fruit (pulp and peel) in the starchy state suitable for the preparation of alcohol or for making into a nourishing bread.

2. Meal in a starchy state from the pulp only for making into a superior kind of bread or porridge.

3. Flakes and meal in a dextrinous state for use in breweries or for making into nourishing soups, puddings, etc.

4. Dried peel and coarse meal prepared from it for feeding cattle and pigs.

5. Banana marmalade.

6. Dried bananas entire without peel put up like dried figs in boxes.

7. Raw alcohol from fresh bananas, and also from dried banana meal.

8. Syrup of bananas for confectionery, for preparations of liquors and for sweetening champagne.

9. Banana meal for the manufacture of glucose.

10. Fibre of banana and plantain prepared from the stems after fruiting, and intended for the manufacture of paper and cordage.

BEGINNING OF A GREAT TRAFFIC

Mr. Asser estimates the entire cost of a ton of banana meal, delivered in Europe, at $23. This includes cost of cultivation, gathering the crop, making the meal, and the freight. At that time the average market value of Indian wheat in Liverpool was $30 per ton. Considering the selling value of the meal to be no greater than that of the wheat, the prices quoted would show a margin of profit equal to about 30 per cent. on the capital invested.

From British Guiana comes the following interesting information about plátano flour, taken from a report by Dr. Shier on the "Starch-producing Plants" of that country:

The plantain is so abundant and cheap that it might, if cut and dried in its green state, be exported with advantage. It is in this unripe state that it is so largely used by the peasantry of this Colony as an article of food. When dried and reduced to the state of meal, it cannot like wheat flour, be manufactured into macaroni or vermicelli, or, at least, the macaroni made from it falls into powder when put into hot water. Plantain meal is prepared by stripping off the husk of the plantain, slicing the core, and drying it in the sun. When thoroughly dry it is powdered and sifted. It has a fragrant odor, acquired in drying, somewhat resembling fresh hay or tea. It is largely employed as the food of infants and invalids. In respect to nutritiveness it deserves a preference over all the pure starches on account of the proteine compounds it contains. The flavor of the meal depends a good deal on the rapidity with which the slices are dried. Above all, the plantain must not be allowed to approach too closely to yellowness or ripeness, otherwise it becomes impossible to dry it. The color of the meal is injured when steel knives are used in husking or slicing, but silver or nickel blades do not injure the color. Full-sized and well-filled bunches give 60 per cent. of core to 40 per cent. of husk and top-stem; but in general it would be found that the core did not much exceed 50 per cent. of dry meal, so that from 20 to 25 per cent. of meal is obtained from the plantain, or 5 pounds from the average bunch of 25 pounds; and an acre of plantain walk of average quality, producing during the year 450 such bunches, would yield 4 tons and 10 pounds of meal.

In 1891, C. W. Meaden wrote from Trinidad to the following effect in relation to a trial shipment of dried bananas:

This experiment will prove of importance to banana growers, as drying bananas seems to open a way no other means offers of utilizing fruit. It overcomes the difficulty of bad roads, long hauls and other drawbacks some planters have to face in marketing bananas.

The result of drying six bunches, weighing an average of 52 pounds per ripe bunch, was 97 pounds of dried fruit. There was a loss of two-thirds in peeling and drying. The fruit sold for $19.40, or 20 cents per pound. Deducting freight charges left $15.47, or a fraction under 16 cents per pound. This was at the rate of $2.72 per bunch. The cost was put at 53 cents, which covered purchase of land, clearing woods and draining, planting, weeding and cutting, drying, fuel, boxes and packing; but did not include cost of dryer, as that would be but a fraction on each bunch dried. After deducting the above there was a profit of $2.19 per bunch.

Mr. Meaden said of this:

I do not desire to set up as a teacher, but facts and figures speak for themselves. The account shown is not an approximate one, but the money has been received and the Canadians are asking for more at the same price. An order is now in hand for 224 pounds for London at 6d. per pound in bulk, the consignee to do the retail packing and advertising. As the fruit is something new it is being sought, and all that can be dried is being profitably disposed of. I may add that the dryer does his work well, turning out the fruit in uniform color. Attention must be paid to this, and also that fruit as nearly as possible of one size be dried, as this facilitates packing. Small ones can be used for stock, etc. Twelve good sized fruits weigh one pound.

The *Daily Gleaner*, of Kingston, Jamaica, said in March, 1899, in reference to an enterprise on the Montpelier estate of Hon. Evelyn Ellis:

As far as dried bananas are concerned the investment is a success. Orders are already taken for more than can be supplied. The factory will be duplicated as soon as possible. Every one who has tasted the bananas is of the opinion that they are superior to figs in every way, and there is likely to be a large home consumption as soon as the factory can supply the market.

Housewives who wish for novelties to lend new charm to their tables, to tickle the palate of the epicure, or to coax the reluctant appetite of the invalid, will find them in novel dainties made from bananas. Excellent and nutritious bread may be made of the flour. Puddings, fritters and sauce have already been mentioned; but bananas glacé are new to most northern folk, and may be made a most delightful addition to our desserts. They are superior to dried figs, for when split into four slices, thickly covered with powdered sugar, and exposed to the sun awhile they turn themselves into a jelly-like, delicious and delicate confection, such as is at its best when made in the native home of the fruit, and packed in pretty boxes to be sent to people of fine taste in the cold North.

Having in view all these facts, why should not multitudes make homes where scorching heat and biting cold are never felt, and tornado and deadly blizzard are unknown; where no destructive floods nor ruinous droughts ever come, and never ceasing winds bring coolness from the sea; where spring is eternal and harvests never end, and delicious fruits yield profusely all the years; where the pine and palm together shade the ground, and the coco and banana yield generous provision for every need; where a little work insures against want and care, and health and leisure make old age secure and content?

Transcriber's Note

Minor typographical errors (i.e. missing punctuation and "egiv" changed to "give") have been corrected.

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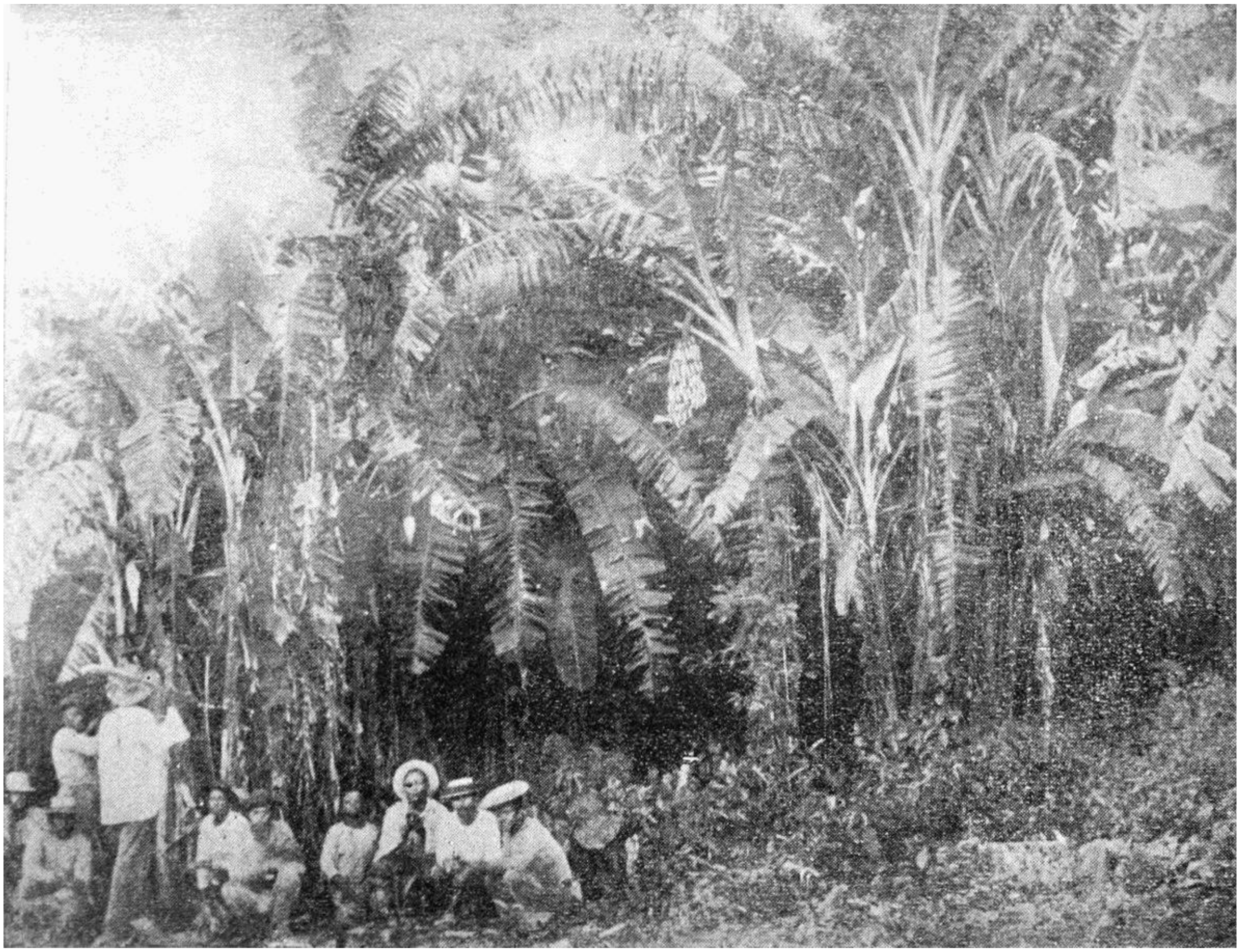

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